Daily Drilling Report
SPD19B-03
9/6/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
41.0	8 1/2"	186.00	72.83	31.70	92.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	224,742.50	1,394.69	10,094,959.55		36.85
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
2,756.00	2,842.00	2,687.71	86.00	20.80	4.1

Operations at Report Time
Drilling @ 2863m.

Operations Summary
Directional drilling 8 1/2" hole section f/ 2756m to 2842m.

Operations next 24hrs
Cont' drilling 8 1/2" hole section.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Sajad Tamaddon	Senior Drilling Sup.
Vahab Kari / Behnam Jahangiri	Night Drilling Sup.
Moslem Yarahmadi	Drilling Engineer
Yaghub Mohammadi	Geologist
GH. R. Babaei	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	9/4/2014	9/26/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	9/6/2014	10/7/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	9/6/2014	10/22/2014
Trip Drill	9/4/2014	10/25/2014

STOP Cards

Number of Reports: 37

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	00:00	24....	8 1/2"	D	03	P	Directional drilling 8 1/2" hole section in rotation and slide mode f/ 2756m to 2842m w/ 500-570 gpm /2200-2400 psi / 20-25 kips/ 50-75 rpm / 15-22 Kft-lbf. Slide Interval 2756m - 2765m Rotation Interval 2765m - 2842m ** Drilling with NL/NG. ** B/R & R/D each stand drilled once. ** Pump 20 bbl Hi-Vis pill when std down. ** Take survey before connection.
00:00	00:00						Sep 07, 2014 00:00-06:00 Cont' Directional drilling 8 1/2" hole section in rotation mode f/ 2842m to 2863m w/ 570 gpm / 2300-2350 psi / 25 kips / 60 rpm / 15-22 Kft-lbf. ** Drilling with NL/NG. ** B/R & R/D each stand drilled once. ** Pump 20 bbl Hi-Vis pill when std down. ** Take survey before connection. >> Last sample @ 2860m: Lst (70 %) + Dol (20 %) + Anhy (10 %)

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
2,842.00	CaCo3/KCL/ Polymer	10.50	54	27.0	22.000	2.6
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.0	0.300	1.500	1.100			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
4	6	0.0	82.0	18.0	33,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
5.000	6.000	8.000	0/100	61.0	1466.0	2745.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
7102.0	140.0	0.0	1.17		8.872876	0.511

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
2.00	6.	300	10.0	996.0	36.7

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Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
Borkan	Supply Boat	9/6/2014 13:45	9/6/2014 16:45	Arrived f/SR1. Received B/L for ABAN 8. Departed to ABAN 8.
Ahmed 6	Supply Boat	9/6/2014 06:00		Delivered DD6 Cargo & B/Loaded Cargo for Kish, Stand by at Loc.
Samarghand	Supply Boat	9/7/2014 02:15	9/7/2014 03:15	Arrived from Kish. Delivered Cargo. Departed to Kish.

Mud Additives

Description	Consumed	Cum Rec	On Loc
Calcium Carbonate Coarse		9.0	8.0
Caustic Soda 2014	5.0	500.0	306.0
Duo-Tec	1.0	185.0	120.0
Inhibitor 303		32.0	32.0
CaCL2 POWDER 94-97%		50.0	50.0
Lime 2014		120.0	120.0
Starch	20.0	1,600.0	397.0
Mica Coarse		80.0	40.0
Mica Fine		160.0	120.0
Mica Medium		80.0	40.0
Nut Plug Fine		90.0	60.0
Nut Plug Medium		60.0	30.0
OYSTER SHELL COARSE		160.0	160.0
OYSTER SHELL XC		40.0	40.0
OYSTER SHELL MEDIUM		40.0	40.0
Quick Seal Coarse		96.0	32.0
Quick Seal Medium		192.0	96.0
SAFE-SCAV CA		40.0	40.0
SAFE-SURF E		4.0	4.0
Soda Ash 25 Kg	26.0	750.0	341.0

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	21.0	20.0	202.0
Cement G	0.0	0.0	92.0
Drill Water	7.0		323.0
Diesel Fuel	11.0		188.0
Barite BB	0.0	0.0	0.0
Barite Bulk	0.0	0.0	129.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
10	8 1/2	Varel	VM713GUX	6006867	PDC	M332	14/14/14/14
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.60	30.90	379,742.40	2,725.00	2,842.00	117.00	3.8	-------
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
500	575	1,825.0	2,360.0	15	27	160	242
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
18.90	27.50	89.00	11.00	435	356	285	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)
2,756.00	2,765.00	2.57		550		2,340.0	27	
2,765.00	2,812.20	8.98	20,000.0	550	50	2,340.0	25	353
2,812.20	2,826.40	4.30	21,000.0	570	60	2,350.0	25	352
2,826.40	2,828.50	0.93	21,000.0	500	60	1,950.0	20	352
2,828.50	2,835.50	1.97	21,000.0	575	60	2,360.0	25	354
2,835.50	2,837.00	0.80	21,000.0	575	75	2,320.0	25	354
2,837.00	2,842.00	1.25	21,000.0	570	60	2,320.0	25	356

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWDP	5 7/8	4.000	27.89	255.95
7" Drilling Jar (COR-3021AD)	7	2.938	5.80	228.06
20 jts 5 7/8" HWDP	5 7/8	4.000	186.35	222.26
X/O	6 1/2	2.813	0.85	35.91
6 3/4" Float sub	6 3/4		0.81	35.06
6 3/4" NMDC	6 3/4	3.250	9.48	34.25
6 3/4" TM - Sub	6 3/4		1.74	24.77
6 3/4" NMDC	6 3/4	3.250	9.48	23.03
8" St. Stabilizer (COR-2491AD)	8	3.000	1.84	13.55
Pony NMDC	6 13/16	3.500	3.09	11.71
6 3/4" Motor (AKO: 1.15° , Sleeve: 8 3/8")	6 3/4		8.34	8.62

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Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
8 1/2" PDC bit	8 1/2		0.28	0.28

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
9/5/2014	9 5/8" Shoe, 2,720.80mKB	Formation Integrity	2,728.00	2,583.00	1,540.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,540.0	5.7	1,540.00		14.00	6,162.4

Directional Survey Data

MD (mKB)	Azm (°)	Incl (°)	TVD (mKB)	NS (m)	EW (m)	VS (m)	DLS (°/30m)	Build (°/30m)
2,787.46	251.70	23.60	2,638.31	-348.92	-768.69	844.17	2.30	2.05
2,816.79	251.70	25.60	2,664.98	-352.75	-780.28	856.32	2.05	2.05
2,845.41	251.40	25.60	2,690.79	-356.67	-792.01	868.62	0.14	0.00

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
S3		2,622.00	2,483.90
S2		2,632.50	2,493.72
S1		2,675.70	2,534.06
Upper Kangan		2,703.00	2,559.14
K1		2,732.00	2,586.77

General Remarks

Comment

>> Requirements: (200 sxs Starch f/ MR #14)

>>> Sent:

** Ahmed 6 to Kish: (PDF 10 x BB of Sodium Chloride) + (Garbage) + (40 sxs Mica Coarse + 40 sxs Mina Medium + 40 sxs Mica Fine + 30 sxs Nut plug Fine + 30 sxs Nut plug M + 64 sxs Q seal C + 96 sxs Q seal M)

** Borkan to aban8: (HPOGC 1 x restrictor jet 42 x 2) + (ABAN equipment)

>>> Received:

** Ahmed 6 f/ Kish: (50 BB Cacl2 + 6 BB CaCo3-C + 32 drums Inhibitor 303 + 160 sxs Oyster Shell C + 40 sxs Oyster Shell M + 120 sxs Lime + 40 sxs Oyster Shell XC + 40 sxs Safe Scav CA + 4 drums Safe Surf E)

** Ahmed 6 f/ Pasargad-100: (PDK Rig up Box)

** Samarghand f/ Kish: (PDK wireline tool basket containing Flexible + TCMRT + Telemetry +Knuckle jnt + CCL + BHC + GEMKO 7" & 9 5/8" + M/U stand + Safety belt + Wooden box + Hole finder + B nose)